

LEONARDO RIBEIRO

<http://leoribeiro.github.io>
Google Scholar

@ leonardofribeiro@gmail.com
in LinkedIn

📍 Seattle, USA

RELEVANT EXPERIENCE

Senior Applied Scientist, Amazon, Apr 2025 - Present 📍 Seattle

- Developed and optimized Amazon Nova models with an emphasis on reasoning, tool use, and RAG performance for multi-step agentic workflows.
- Built scalable data-generation and evaluation pipelines for tool-augmented agents (search, web browsing, and code execution), supporting use cases such as multi-hop QA, research/synthesis, troubleshooting, and code-assisted analysis; validated on benchmarks including GAIA, Humanity's Last Exam, and BrowseComp.
- Designed end-to-end RL training environments for single- and multi-agent LLMs, and developed reward functions and data recipes (e.g., GRPO) for outcome-based optimization in multi-turn tool use.

Applied Scientist, Amazon, Oct 2022 - Apr 2025 📍 Seattle

- Developed and post-trained large-scale Amazon Nova 1.0 LLMs (SFT, RLHF) for RAG and multimodal RAG, and built automated evaluation workflows to measure quality and reliability.
- Built scalable post-training data pipelines, including (i) DPO preference pairs using LLM-as-a-judge with grounding / faithfulness checks and (ii) multimodal SFT data for InfoSeek-style QA that requires retrieval-augmented answering.

Researcher, Technische Universität Darmstadt, Nov 2018 - Sep 2022 📍 Darmstadt, Germany

- My research focused on neural graph methods for natural language generation.
- Supervised bachelor's and master's theses; evaluated candidates for PhD and postdoctoral positions.

Software Engineer, Labor Court of Rio de Janeiro, 2015 - 2018 📍 Rio de Janeiro, Brazil

- Engineered core features for the Electronic Judicial Process, a system used by millions of users to manage and track legal proceedings.

Software Engineer, Labor Court of Minas Gerais, 2013 - 2015 📍 Belo Horizonte, Brazil

- Designed and implemented systems for the Federal Justice.

EDUCATION

PhD in Computer Science, Technische Universität Darmstadt, 2018 - 2022 📍 Darmstadt, Germany

- **Dissertation:** [Graph-based Approaches to Text Generation](#). Advisor: Iryna Gurevych. Grade: *summa cum laude*.

MSc in Computer Engineering, Universidade Federal do Rio de Janeiro, 2015 - 2017 📍 Rio de Janeiro, Brazil

- **Thesis:** [Learning Network Node Representations from Structural Identity](#). Advisor: Daniel Ratton Figueiredo. Recipient of the 2018 Brazilian Artificial and Computational Intelligence Thesis Award.

BSc in Computer Engineering, Centro Federal Educ. Tec. de MG, 2009 - 2014 📍 Timóteo, Brazil

- Silver Medal (2nd in class).

TECHNICAL REPORTS

- [Amazon Nova 2: Multimodal reasoning and generation models](#), [Amazon Technical Reports](#) Dec. 2025. Core Contributor.
- [The Amazon Nova Family of Models: Technical Report and Model Card](#) Dec. 2025. Core Contributor.

SELECTED PUBLICATIONS

- [RefusalBench: Generative Evaluation of Selective Refusal in Grounded Language Models](#). Aashiq Muhamed, Leonardo FR Ribeiro, Markus Dreyer, Virginia Smith, Mona T Diab. EACL 2026.
- [Deep research comparator: A platform for fine-grained human annotations of deep research agents](#). Prahaladh Chandrasan, Jiahe Jin, Zhihan Zhang, Tevin Wang, Andy Tang, Lucy Mo, Morteza Ziyadi, Leonardo F. R. Ribeiro, Zimeng Qiu, Markus Dreyer, Akari Asai, Chenyan Xiong. WWW 2026 (Demo).

- **RAGferee: Building Contextual Reward Models for Retrieval-Augmented Generation.** Andrei C Coman, Ionut-Teodor Sorodoc, Leonardo FR Ribeiro, Bill Byrne, James Henderson, Adrià de Gispert. EMNLP 2025.
- **XRAG: Cross-lingual Retrieval-Augmented Generation.** Wei Liu, Sony Trenous, Leonardo FR Ribeiro, Bill Byrne, Felix Hieber. EMNLP 2025 (Findings)
- **GaRAGE: A Benchmark with Grounding Annotations for RAG Evaluation.** Ionut-Teodor Sorodoc, Leonardo F. R. Ribeiro, Rexhina Blloshmi, Christopher Davis, Adrià de Gispert. ACL 2025 (Findings)
- **Learning When to Retrieve, What to Rewrite, and How to Respond in Conversational QA.** Nirmal Roy, Leonardo F. R. Ribeiro, Rexhina Blloshmi, Kevin Small. EMNLP 2024 (Findings).
- **Speechworthy Instruction-tuned Language Models.** Hyundong Cho, Nicolaas Jedema, Leonardo F. R. Ribeiro, Karishma Sharma, Pedro Szekely, Alessandro Moschitti, Ruben Janssen, and Jonathan May. EMNLP 2024.
- **REFINESUMM: Self-Refining MLLM for Generating a Multimodal Summarization Dataset.** Vaidehi Patil, Leonardo F. R. Ribeiro, Mengwen Liu, Mohit Bansal, Markus Dreyer. ACL 2024.
- **Generating Summaries with Controllable Readability Levels.** Leonardo F. R. Ribeiro, Mohit Bansal, Markus Dreyer. EMNLP 2023.
- **FactGraph: Evaluating Factuality in Summarization with Semantic Graph Representations.** Leonardo F. R. Ribeiro, Mengwen Liu, Iryna Gurevych, Markus Dreyer, Mohit Bansal. NAACL 2022.
- **Structural Adapters in Pretrained Language Models for AMR-to-Text Generation.** Leonardo F. R. Ribeiro, Yue Zhang and Iryna Gurevych, EMNLP 2021.
- **Investigating Pretrained Language Models for Graph-to-Text Generation.** Leonardo F. R. Ribeiro, Martin Schmitt, Hinrich Schütze and Iryna Gurevych. NLP for Conversational AI, EMNLP 2021.
- **Modeling Global and Local Node Contexts for Text Generation from Knowledge Graphs.** Leonardo F. R. Ribeiro, Yue Zhang, Claire Gardent and Iryna Gurevych. Transactions of the Association for Computational Linguistics (TACL), 2020.
- **Enhancing AMR-to-Text Generation with Dual Graph Representations.** Leonardo F. R. Ribeiro, Claire Gardent and Iryna Gurevych. 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP).
- **Ranking Generated Summaries by Correctness: An Interesting but Challenging Application for Natural Language Inference.** Tobias Falke, Leonardo F. R. Ribeiro, Prasetya Utama, Ido Dagan and Iryna Gurevych. 57th Annual Meeting of the Association for Computational Linguistics (ACL), 2019.
- **struc2vec: Learning Node Representations from Structural Identity.** Leonardo F. R. Ribeiro, Pedro H. P. Saverese, Daniel R. Figueiredo. ACM SIGKDD 2017.

ACADEMIC SERVICE

- **Senior Area Chair** – NAACL 2025 (2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics).
- **Tutorial Chair** – ACL 2024 (62nd Annual Meeting of the Association for Computational Linguistics).
- **Area Chair** – ACL Rolling Review: COLING 2022; EACL 2024; NAACL 2024; ACL 2024. Tracks: Summarization, Generation, Machine Learning for NLP.
- **Program Committee / Reviewer** – Reviewed 100+ papers for ARR; EMNLP 2020–2023; ACL 2021–2023; NAACL 2021; AAAI 2021–2024; EACL 2023; and workshops at ACL/EACL/NAACL/EMNLP/AAAI, as well as JMLR.

AWARDS & FELLOWSHIPS

- 2018 – **Best Master's Thesis Award.** 11th Thesis and Dissertation on Artificial and Computational Intelligence Contest (Brazil).
- 2012–2013 – **Science Without Borders Scholarship** (Brazilian government), one-year study at Universidad del País Vasco (Spain).
- 2016 – **Honorable Mention** for a paper at the Brazilian Workshop on Social Network Analysis and Mining (BraSNAM).
- 2014 – **Silver Medal** (2nd in class), BSc in Computer Engineering.
- 2008 – **Gold Medal** (1st in class), Technical Degree in Informatics.

SKILLS

- Python, PyTorch; RAG; tool-use agents (search/browse/code); evaluation & benchmarking; post-training (SFT, RLHF, DPO); reward modeling / GRPO. vLLM; DeepSpeed; Hugging Face Datasets; Arrow/Parquet; AWS (EKS); Docker; Kubernetes; distributed training/inference; Git; Linux; Kiro (agentic coding assistant).
- **Languages:** Portuguese (native), English (fluent), Spanish (intermediate), German (basic).